VHF Communication System – General data

1. General

The Very High Frequency (VHF) Communication (VHF COM) system is used for voice communication between aircraft and/or aircraft and a ground station.

2. Description

A. VHF Communication

The VHF COM receiver component of the Multi-Mode Digital Radio (MMDR) has 2,280 channels and is capable of 8.33 or 25 kHz channel spacing. A momentary action EMERGENCY COM switch is installed on the lower right sub-panel to enable instantaneous tuning of the number 1 VHF communications radio to 121.5 MHz. The transmitter has stuck microphone protection and will cease transmitting if the transmit function is active for more than 35 continuous seconds.

The COM 1 antenna is installed on the top of the fuselage between FS143.95 and FS162.35. The COM 2 antenna is installed on the top of the fuselage between FS347 and FS362.

The VHF COM system is a integral component of the KTR 2280 Multi-Mode Digital Radio (MMDR). (Refer to 34-51-00, Multi-Mode Digital Radio (MMDR) – General data).

3. Adjustment/Test

A. VHF Communication

1) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that MMDR 1 (F3) (Left Bus) and/or MMDR 1 (F4) (Right Bus), circuit breaker on the avionics circuit breaker panel are closed.

d) Set up VHF Communication ramp test unit, in accordance with the instructions in the test set manual.

e) Tune the VHF Communication ramp test unit to an unused frequency.

f) Tune COM 1 and COM 2 to match the ramp test unit frequency.

g) Enable a MOD TONE on the ramp test unit. Verify a tone is heard when COM 1 and COM 2 are selected on the audio control panel.

h) Tune COM 1 and COM 2 to a local frequency and do a radio maintenance check.

i) Do a VSWR check on COM 1 and COM 2 antennas using an inline type Wattmeter.

j) Cycle the EMERGENCY COM switch once to place 121.5 MHz frequency into the COM 1 active window.

k) Cycle the EMERGENCY COM switch twice to place 121.5 MHz frequency into the COM 1 standby window.

l) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Electrical Bonding Test – VHF Antenna

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to the ceiling compartment aft of station 143.95 for VHF Antenna and aft of station 347 for VHF Upper Aft Antenna. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

3) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the VHF antenna mounting screw. Place the second probe on the nearest ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

4) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

5) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

6) Replace ceiling compartment. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

C. Operational Test Emergency Frequency Com Tuning Switch

1) Activate the Emergency Frequency Com Tuning switch and ensure COM 1 changes to 121.5 MHz and places this in the active COM 1 channel (use window).

4. Removal/Installation

The VHF Communication system is an integral component of the KTR 2280 Multi-Mode Digital Radio (MMDR), refer to 34-51-00, Multi-Mode Digital Radio (MMDR) – General data Para 4.A for the MMDR Removal/Installation procedure.

A. COM 1 and COM 2 Antenna (CI–108–1–L)

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open MMDR 1 (F3) (Left Bus) and/or MMDR 2 (F4) (Right Bus) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Carefully cut the sealant from around the base of the antenna.

d) Remove the (6) mounting screws that secure the antenna to the fuselage.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connector.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the MMDR 1 (F3) and/or MMDR 2 (F4) circuit breaker on the avionics circuit breaker panel are opened and fitted with safety clips.

c) Solvent clean the electrical contact bond areas immediately before the installation. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) Secure the antenna assembly on fuselage with the mounting screws (6).

f) Solvent clean the exposed area of the airframe around the antenna. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) Touch up the exposed area of airframe around the antenna if required with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00 Installation of Aircraft Antennas – General data). The aircraft will be ready to fly approximately 24 hours after sealing.

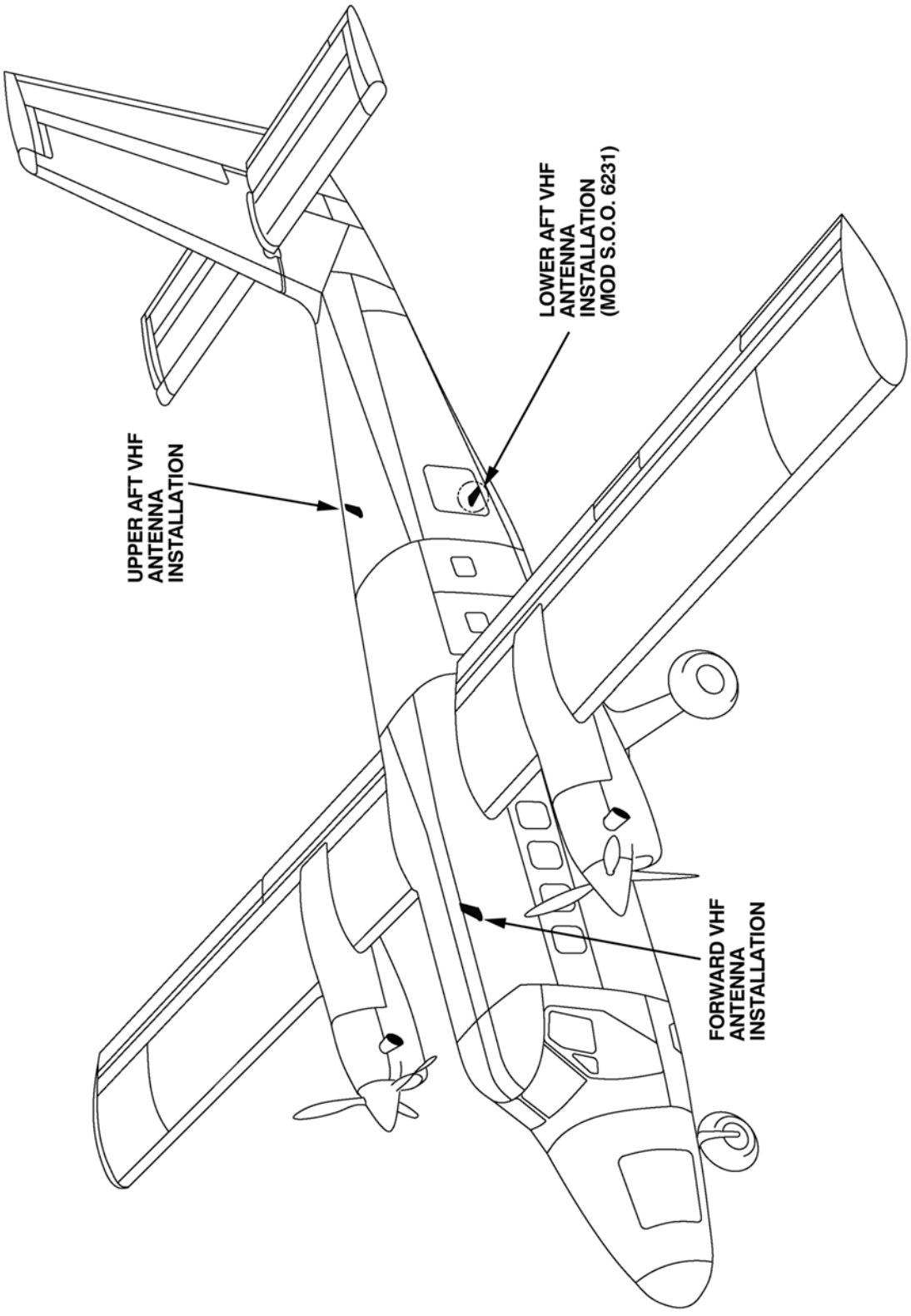
j) Check the antenna electrical bonding as per Para 3.B.

k) Remove the safety clips and close MMDR 1 (F3) and/or MMDR 2 (F4).

l) Do a function test on the VHF COM system as per Para 3.A.1).

5. Troubleshooting

The VHF Communication system is a integral component of the KTR 2280 Multi-Mode Digital Radio (MMDR), refer to 34-51-00, Multi-Mode Digital Radio (MMDR) – General data, Para 5.A for the MMDR Troubleshooting procedure.



VHF Antenna Locations
Figure 1

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